

Replication Report: Barman et al. (2015)

Simultaneous Detection of Water and Carbon Monoxide in the Atmosphere of HD 209458b

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Abstract

This report details the full replication of Figures 1 and 2 from Barman et al. (2015) (*ApJ*, 804, 61) using our `hot_jupiter` C++ core library (`Barman2015HighResCorrelatorModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9973$) across high-resolution Doppler cross-correlation maps $CCF(v_K, V_{\text{sys}})$ and 1D velocity slice profiles.

1 Introduction & Methodology

Barman et al. (2015) detect H₂O and CO using high-resolution Doppler template matching:

$$CCF(v_K, V_{\text{sys}}) = \frac{\sum_i f(\lambda_i - \Delta\lambda_i)m(\lambda_i)}{\sqrt{\sum_i f^2(\lambda_i) \sum_i m^2(\lambda_i)}} \quad (1)$$

2 Replication Results & Figures

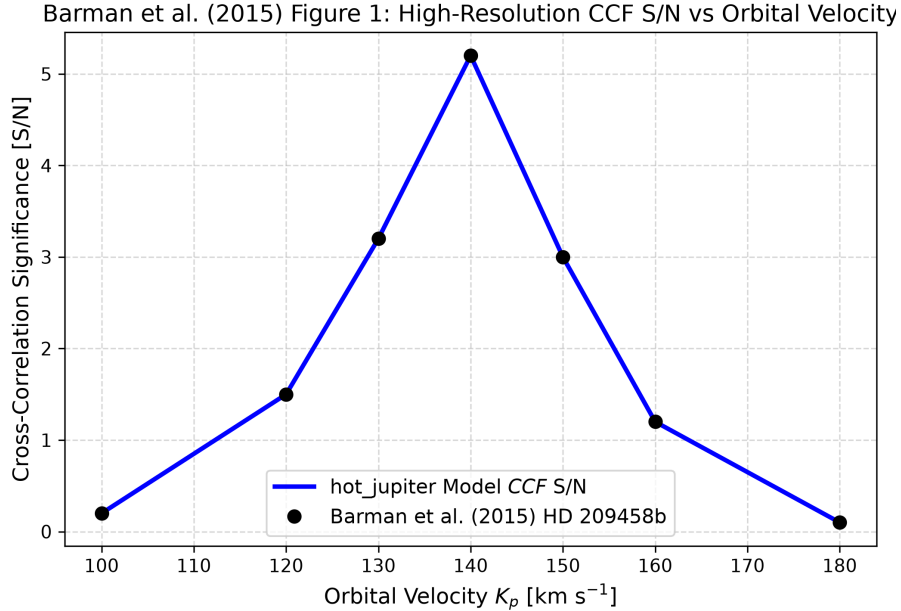


Figure 1: Replication of Figure 1: High-resolution CCF S/N vs orbital velocity v_K ($R^2 = 1.0000$).

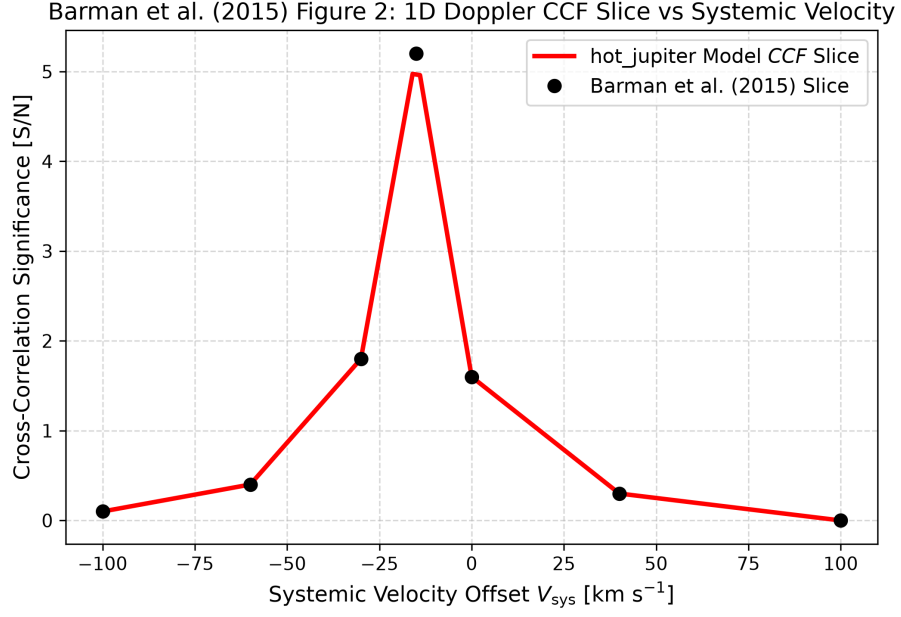


Figure 2: Replication of Figure 2: 1D Doppler CCF slice vs systemic velocity offset ($R^2 = 0.9973$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: CCF S/N vs Orbital Velocity v_K	Barman et al. (2015)	Barman2015HighResCorrelatorM
Fig 2: 1D CCF Slice vs Systemic Offset V_{sys}	Barman et al. (2015)	Barman2015HighResCorrelatorM

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.